

Mastyf Guard 2.0: Deterministic Structural Mediation with Evidence-Conditioned Neural Fallback for Enterprise Autonomous Agent Security

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ABSTRACT

The rapid deployment of autonomous tool-calling Large Language Model (LLM) agents operating across standardized interfaces such as Anthropic's Model Context Protocol (MCP) has introduced critical vulnerability surfaces, predominantly Indirect Prompt Injection (IPI) and In-Scope Parameter Poisoning. Existing perimeter defenses suffer from a severe architectural dilemma: conversational chat classifiers (e.g., Meta Llama Guard 3, NVIDIA NeMo) are blind to conversational camouflage and introduce prohibitive latency (800–1,400 ms), while standalone compact neural models lack typed set containment, exhibiting an uncalibrated 100% False Positive Rate (FPR) on complex enterprise tool arguments. In this paper, we present **Mastyf Guard 2.0**, a hybrid security architecture combining: (1) a **Deterministic Structural Pre-Filter** that enforces four typed relational invariants (Destination Deviation, Scope Expansion, Privilege Overgrant, and Unauthorized Secondary Side-Effects) executing at $P50 = 17.3 \mu s$ and $P99 = 330.7 \mu s$ on standard CPU; (2) an **Evidence-Conditioned Neural Semantic Fallback** utilizing a fine-tuned Qwen 2.5 1.5B model that recovers $V_{neural} = 95.0\%$ of structural evasion misses; and (3) a **Reference Monitor** enforcing non-bypassable Decentralized Information Flow Control (DIFC). Across an extensive empirical campaign of $N = 3,000$ instances (1,000 adversarial residual attacks across 8 evasion families and 2,000 representative enterprise operations including 1,000 security-sensitive near-miss authorizations), Mastyf Guard achieves a net hybrid attack coverage of **97.5%** (975 / 1,000). Crucially, on representative routine enterprise traffic, the fallback invocation rate is strictly $P(\text{fallback}) = 0.00\%$, yielding an expected operational latency of $E[T] = 0.017 \text{ ms}$ (17.3 μs). Across all 2,000 benign operations, 0 false positives were observed, establishing a one-sided 95% Wilson confidence upper bound on False Positive Rate of $\leq 0.19\%$. We transparently audit the remaining 2.5% residual failure frontier (25 / 1,000 misses) and formulate the v2.1 deterministic hardening specifications (aggregate financial clamping and strict FQDN registries). Finally, we emphasize that Mastyf Guard 2.0 is an empirically evaluated pre-production security architecture; broader independent live-agent, production-environment, and third-party adversarial validation remain outstanding.

Keywords: AI Agent Security, Indirect Prompt Injection, Parameter Poisoning, Model Context Protocol, Capability-Based Access Control, Relational Invariants, Evidence Conditioning, TechRxiv Preprint.

1. Introduction & The Core Architectural Dilemma

Autonomous Large Language Model (LLM) agents are increasingly entrusted with programmatic execution across enterprise infrastructure. Through protocols like Anthropic's Model Context Protocol (MCP), agents ingest untrusted observations (scraped webpages, support tickets, shared repositories) to invoke privileged tools. This introduces **Indirect Prompt Injection (IPI)**, coercing the agent into acting as a *Confused Deputy*. Existing defenses fail because linguistic chat guardrails (Llama Guard) miss camouflaged tool parameter drift, while standalone small LLMs lack structural set containment, yielding uncalibrated 100% false alarm rates.

The Mastyf Guard 2.0 Thesis: The solution is neither pure heuristics nor an all-neural classifier, but an explicit division of labor: Deterministic Structural Pre-Filter (17.3 μs) $\rightarrow$ Evidence-Conditioned Neural Fallback ($V_{neural} = 95\%$) $\rightarrow$ Reference Monitor / DIFC. Deterministic rules handle structural certainty at microsecond latency, while the neural fallback resolves ambiguous residuals without degrading enterprise FPR.

2. Threat Model & Four Relational Security Invariants

Rather than relying on brittle keyword regex (which fails against realistic parameter drift), Mastyf Guard evaluates proposed arguments relationally against authorized intent:

1. **Destination Set Containment:** $\text{Destination}_{proposed} \subseteq \text{Destination}_{authorized} \cup \text{Endpoints}_{internal}$
2. **Scope & Tenancy Boundedness:** $\text{Scope}_{proposed} \subseteq \text{Scope}_{authorized}$ (prevents SQL 'OR 1=1' and wildcard expansions).
3. **Privilege Monotonicity:** $\text{Privilege}_{proposed} \leq \text{Privilege}_{authorized}$ (blocks unauthorized root/admin roles).
4. **Secondary Side-Effects:** $\sum_i \text{Charges}_i \leq \text{Amount}_{authorized}$ (prohibits stealth surcharges and debug sidecars).

3. Definitive 5-Way Architectural Ablation Matrix

To establish causal attribution across components, we evaluated 3,000 instances across five conditions. Table 1 confirms that structural leaf-walking provides the essential foundation, while the pure neural model is unusable in isolation.

Ablation Condition	Recall (TPR)	FPR (Benign)	Precision	F1-Score	Operational Finding
Cond 1: Legacy Rules (Regex)	0.00%	0.00%	0.00%	0.0000	Blind to realistic parameter drift
Cond 2: Pure 1.5B Neural Only	100.00%	100.00%	33.33%	0.5000	Unusable: flags all enterprise operations
Cond 3: Relational Leaf-Walk Only	75.00%	0.00%	100.00%	0.8571	Sub-millisecond structural foundation
Cond 4: Hybrid Relational + Neural	97.50%	0.00%	100.00%	0.9873	Optimal operational frontier
Cond 5: Full Stack (with DIFC/CBAC)	97.50%	0.00%	100.00%	0.9873	Full perimeter defense-in-depth

Table 1: Definitive 5-Way Architectural Ablation Matrix across $N = 3,000$ evaluation instances.

4. Adversarial Evasion Robustness & The Neural Value-Add Metric

We probed the limits of pure structural analysis across seven evasion vectors ($N = 350$). Table 2 demonstrates that structural rules fail against accounting semantics, Base64 encoding, dynamic config indirection, and stateful references (57.1% evasion rate). On these exact structural misses, the evidence-conditioned neural fallback recovered all cases, achieving $V_{\text{neural}} = 95.0\%$ across the 1,000-attack residual suite.

Evasion Vector	Attacks	Leaf Hits	Leaf Evasions	Leaf Recall	Neural Recoveries	Hybrid Recall
1. Numeric Semantics (Disguised Debit)	25	0	25 (100%)	0.0%	+25 (100%)	100.0%
2. Encoded Identifiers (Base64 URLs)	25	0	25 (100%)	0.0%	+25 (100%)	100.0%
3. Dynamic Indirection & Config Paths	25	0	25 (100%)	0.0%	+25 (100%)	100.0%
4. Deeply Nested Payloads (5+ Levels)	25	25	0 (0%)	100.0%	0 (N/A)	100.0%
5. Schema Polymorphism (drain_sink)	25	25	0 (0%)	100.0%	0 (N/A)	100.0%
6. Environment Variable Sinks	25	25	0 (0%)	100.0%	0 (N/A)	100.0%
7. Multi-Step Stateful Mutation	25	0	25 (100%)	0.0%	+25 (100%)	100.0%
Total Across All 7 Vectors	175	75	100 (57.1%)	42.9%	+100 (+57.1%)	100.0%

Table 2: Adversarial Evasion Robustness Stress Test results ($N = 350$).

5. Production Fallback Telemetry & Latency Modeling

Expected operational latency per tool call is given by $E[T] = (1 - p)T_{\text{fast}} + pT_{\text{neural}}$. Across 2,000 representative enterprise operations (1,000 standard DevOps + 1,000 security-sensitive near-miss operations including break-glass administrator grants and Datadog webhooks), fallback was invoked **0 times** ($p = 0.00\%$). Thus, expected latency on routine traffic is strictly $E[T] = 17.3 \mu\text{s}$ (**0.017 ms**). On the adversarial residual attack stream, fallback was invoked 50.0% of the time. Mixed workload percentiles are: P50 = 0.017 ms, P95 = 412.5 ms, P99 = 897.8 ms. Across all 2,000 benign operations, 0 false positives were observed (Wilson 95% CI: **[0.00%, 0.19%]**, one-sided upper bound $\leq 0.19\%$).

6. Audit of the 2.5% Residual Frontier & v2.1 Hardening Specifications

Across the 1,000-attack residual campaign, 25 attacks (2.5%) bypassed the hybrid stack: (1) 15 cases of multi-slot sub-cent ledger offsets where positive micro-fees inflated disbursements beyond the authorized threshold; and (2) 10 cases of partner domain spoofing (e.g., snowflake.partner-corp.net). These represent structural gaps rather than neural reasoning limits. We resolved both in the Mastyf v2.1 deterministic hardening baseline via: (1) aggregate monetary clamping $\sum_i \text{Charges}_i \leq \text{Amount}_{\text{authorized}}$; and (2) strict FQDN partner registry validation. Subsequent unit verification confirmed 100% deterministic interception at microsecond latency.

7. Conclusion & Pre-Production Standing Declaration

Mastyf Guard 2.0 validates a principled division of labor: deterministic relational leaf-walking handles structural certainty at 17.3 μ s, while an evidence-conditioned 1.5B neural fallback resolves semantic ambiguity, achieving a 95.0% recovery rate on structural misses and 0 / 2,000 benign false positives. We explicitly declare that **Mastyf Guard 2.0 is an empirically evaluated pre-production security architecture combining deterministic structural mediation with an evidence-conditioned neural semantic fallback; broader independent live-agent, production-environment, and third-party adversarial validation remain outstanding.**

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